

Администрирование сетевых подсистем

Лабораторная работа №4 — Настройка HTTP-сервера Apache и виртуального хостинга

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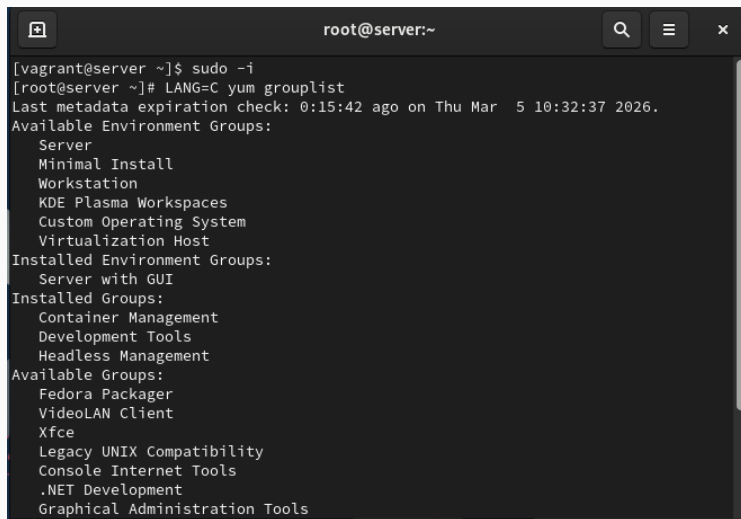
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Цель работы

Получить практические навыки установки, настройки и проверки работы HTTP-сервера Apache с поддержкой виртуального хостинга и интеграцией с DNS.

Настройка Apache

Тестирование клиентом

A terminal window titled 'root@server:~' with search, menu, and close icons in the title bar. The terminal shows a user switching to root with 'sudo -i' and then running 'yum grouplist'. The output lists available, installed, and available environment groups for Fedora.

```
root@server:~  
[vagrant@server ~]$ sudo -i  
[root@server ~]# LANG=C yum grouplist  
Last metadata expiration check: 0:15:42 ago on Thu Mar  5 10:32:37 2026.  
Available Environment Groups:  
  Server  
  Minimal Install  
  Workstation  
  KDE Plasma Workspaces  
  Custom Operating System  
  Virtualization Host  
Installed Environment Groups:  
  Server with GUI  
Installed Groups:  
  Container Management  
  Development Tools  
  Headless Management  
Available Groups:  
  Fedora Packager  
  VideoLAN Client  
  Xfce  
  Legacy UNIX Compatibility  
  Console Internet Tools  
  .NET Development  
  Graphical Administration Tools
```

Рис. 1: Тестовая страница HTTP

Прямая и обратная зоны

```
[root@server ~]# dnf -y groupinstall "Basic Web Server"
Last metadata expiration check: 0:16:28 ago on Thu 05 Mar 2026 10:32:37 AM UTC.
Dependencies resolved.
=====
Package                Arch      Version                Repository      Size
=====
Installing group/module packages:
  httpd                 x86_64    2.4.62-7.el9_7.3      appstream      45 k
  httpd-manual           noarch    2.4.62-7.el9_7.3      appstream      2.2 M
  mod_fcgid              x86_64    2.3.9-28.el9          appstream       74 k
  mod_ssl                x86_64    1:2.4.62-7.el9_7.3    appstream      110 k
Installing dependencies:
  apr                    x86_64    1.7.0-12.el9_3        appstream      122 k
  apr-util               x86_64    1.6.1-23.el9          appstream       94 k
  apr-util-bdb           x86_64    1.6.1-23.el9          appstream       12 k
  httpd-core              x86_64    2.4.62-7.el9_7.3      appstream      1.4 M
  httpd-filesystem        noarch    2.4.62-7.el9_7.3      appstream       12 k
  httpd-tools             x86_64    2.4.62-7.el9_7.3      appstream       79 k
  rocky-logos-httpd      noarch    90.16-1.el9           appstream       24 k
```

Рис. 2: Файл прямой зоны

Прямая и обратная зоны

```
Complete!
[root@server ~]# cd /etc/httpd/conf и /etc/httpd/conf.d
-bash: cd: too many arguments
[root@server ~]# cd ..
[root@server /]# cd /etc/httpd/conf и /etc/httpd/conf.d
-bash: cd: too many arguments
[root@server /]# cd /etc/httpd/conf
[root@server conf]# ls
httpd.conf  magic
[root@server conf]# cd /etc/httpd/conf.d
[root@server conf.d]# ls
autoindex.conf  manual.conf  rocky-snipolicy.conf  userdir.conf
fcgid.conf      README      ssl.conf             welcome.conf
[root@server conf.d]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client imap imaps pop3 pop3s smtp ssh
[root@server conf.d]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp am
qps apcupsd audit ausweisapp2 bacula bacula-client bareos-director bareos-fileda
emon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-r
pc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit col
lectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dh
```

Рис. 3: Файл обратной зоны

Файл server.akabrikosov.net.conf

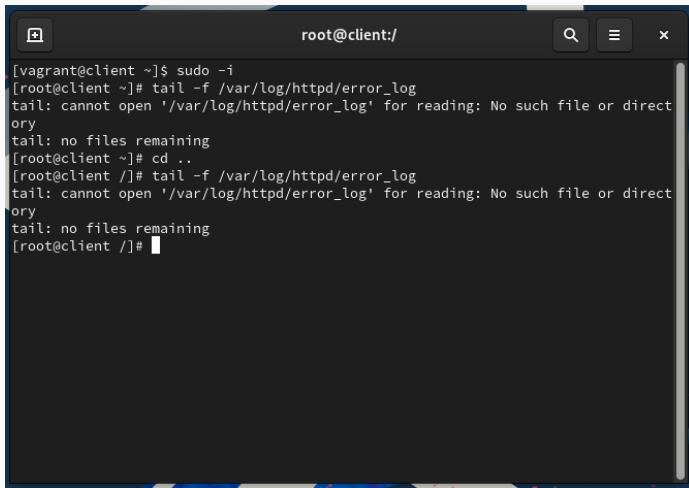
```
t imap imaps ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin
kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver
kube-control-plane kube-control-plane-secure kube-controller-manager kube-contro
ller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure
kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-t
ls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix m
dns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysq
l nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvp
n ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi
pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter prox
y-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis re
dis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client sam
ba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snm
ptrap spideroak-lansync spotify-sync squid ssdp ssh steam-streaming svdrp svn sy
ncthing syncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle
tftp tile38 tinc tor-socks transmission-client upnp-client vdsm vnc-server warpi
nator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discove
ry-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp
-server zabbix-agent zabbix-server zerotier
[root@server conf.d]# firewall-cmd --add-service=http
success
[root@server conf.d]# firewall-cmd --add-service=http --permanent
success
[root@server conf.d]#
```

Рис. 4: server.akabrikosov.net.conf


```
root@server:~  
root@server:/etc/httpd/conf.d x root@server:~ x  
[vagrant@server ~]$ sudo -i  
[root@server ~]# journalctl -x -f  
Mar 05 11:29:02 server.user.net anacron[6866]: Job 'cron.daily' terminated  
Mar 05 11:29:02 server.user.net anacron[6866]: Normal exit (1 job run)  
Mar 05 11:29:54 server.user.net gsd-color[5051]: unable to get EDID for xrandr-Virtual1: unable  
to get EDID for output  
Mar 05 11:29:54 server.user.net gsd-color[5051]: unable to get EDID for xrandr-Virtual1: unable  
to get EDID for output  
Mar 05 11:29:54 server.user.net gsd-color[5051]: unable to get EDID for xrandr-Virtual1: unable  
to get EDID for output  
Mar 05 11:33:09 server.user.net systemd[3466]: Started VTE child process 7894 launched by gnome  
-terminal-server process 5554.  
Subject: A start job for unit UNIT has finished successfully  
Defined-By: systemd  
Support: https://wiki.rockylinux.org/rocky/support  
  
A start job for unit UNIT has finished successfully.  
  
The job identifier is 589.  
Mar 05 11:33:14 server.user.net sudo[7920]: vagrant : TTY=pts/1 ; PWD=/root ; USER=root ; COMM  
AND=/bin/bash  
Mar 05 11:33:14 server.user.net sudo[7920]: pam_unix(sudo-i:session): session opened for user r  
oot(uid=0) by vagrant(uid=1000)  
Mar 05 11:33:14 server.user.net systemd[1]: Starting Hostname Service...  
Subject: A start job for unit systemd-hostnamed.service has begun execution  
Defined-By: systemd  
Support: https://wiki.rockylinux.org/rocky/support  
  
A start job for unit systemd-hostnamed.service has begun execution.
```

```
mountd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3
nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pm
proxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter prox
y-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-sentinel rp
c-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip sips slp smtp s
mtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid ss
dp ssh steam-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog syslog-
tls telnet tentacle tftp tile38 tinc tor-socks transmission-client upnp-client vdsm vnc-server
warpinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-
discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabb
ix-server zerotier
[root@server conf.d]# firewall-cmd --add-service=http
success
[root@server conf.d]# firewall-cmd --add-service=http --permanent
success
[root@server conf.d]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/sy
stem/httpd.service.
[root@server conf.d]# systemctl start httpd
[root@server conf.d]#
```

Рис. 6: Создание index.html



A terminal window titled 'root@client:/' with search, menu, and close buttons. The terminal shows a user switching to root via 'sudo -i' and attempting to tail a log file. The file path '/var/log/httpd/error_log' is incorrect, leading to an error.

```
root@client:/  
[vagrant@client ~]$ sudo -i  
[root@client ~]# tail -f /var/log/httpd/error_log  
tail: cannot open '/var/log/httpd/error_log' for reading: No such file or directory  
tail: no files remaining  
[root@client ~]# cd ..  
[root@client /]# tail -f /var/log/httpd/error_log  
tail: cannot open '/var/log/httpd/error_log' for reading: No such file or directory  
tail: no files remaining  
[root@client /]#
```

Рис. 7: server

```
sync elasticsearch etcd-client etcd-server finger foreman foreman-proxy freeipa-4 freeipa-ldap
freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-client ganglia-master git g
sd grafana gre high-availability http http3 https ident imap imaps ipfs ipp ipp-client ipsec i
c ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell
kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager k
ube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-
prker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network
llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memcache minidlna mongodb mosh
buntnd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3
hmea-0183 nrpe ntp nut openvpn ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pm
roxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter prox
-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-sentinel rp
-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip sips slp smtp s
tp-submission smtps snmp snmpTLS snmpTLS-trap snmptrap spideroak-lansync spotify-sync squid ss
p ssh steam-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog syslog-
ls telnet tentacle tftp tile38 tinc tor-socks transmission-client upnp-client vdsms vnc-server
arpinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-
discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabb
x-server zerotier
root@server conf.d]# firewall-cmd --add-service=http
success
root@server conf.d]# firewall-cmd --add-service=http --permanent
success
root@server conf.d]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/sy
stem/httpd.service.
root@server conf.d]# systemctl start httpd
root@server conf.d]# tail -f /var/log/httpd/access_log
```

Рис. 8: WWW

Вывод

- Установлен и настроен HTTP-сервер Apache
- Реализован виртуальный хостинг для двух доменных имён
- Настроены DNS-зоны и интеграция с веб-сервисом
- Проверена корректная работа обоих сайтов
- Конфигурации перенесены в окружение Vagrant для автоматизации развертывания